

Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science

Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Research

Undergraduate Researcher, Abbasi Lab, UCSF

Sep 2026 – Present

- Applying **unsupervised and semi-supervised deep learning** (deep autoencoders, non-negative matrix factorization (NMF), semi-supervised CNNs) to structural MRI/fMRI from thousands of UCSF patients and public datasets, plus Allen Institute gene expression images, to relate patterns in unlabeled medical images to patient health outcomes.
- Applying **large language model (LLM) interpretability methods** to evaluate the validity of outputs from spatial queries in spatial transcriptomics, testing on smaller datasets and on larger spatial foundation models.

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA

Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using Levenshtein distance over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus (~1M **annotated words across ~10K texts**).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia

Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Trained a CV pipeline for **real-time pile depth tracking** — object detection to localize the pile, YOLO segmentation and line detection to track painted depth markings, an open-source digit detector to identify line numbers, and audio-visual synchronization to isolate each hammer strike, converting pixel-space line displacement into inch/foot measurements via a depth map.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.
- **Helped file a provisional patent**, designing a tripod-mounted 1000fps camera and Jetson Orin GPU capture rig.

Projects

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3)

github.com/FavouritePlayer/ant_sim

- Trained PPO quadruped policies in MuJoCo/Gymnasium, benchmarking flat-ground baselines against terrain-adapted and damage-robust variants across **10 matched random seeds per condition**.
- Terrain adaptation: policy trained on randomized heightfields **achieves 2.1x the reward** (893 vs. 424) and 40% lower fall rate than the flat-trained baseline on unseen rough terrain.
- Leg-damage robustness: a specialist policy trained with one leg permanently amputated reaches ~49x the flat-baseline reward on that leg, but **does not transfer to other legs** (100% fall rate).
- Built a compound policy router that switches checkpoints by damaged leg, **beating any single policy** (836 vs. 497 mean reward) across a 4-leg × 10-seed evaluation sweep.
- Validated all headline results via **multi-seed training replication and an automated regression test suite**; generated comparison videos for each experiment.

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy)

github.com/FavouritePlayer/face-morpher

- Implemented affine warping, inverse mapping, and bilinear interpolation **from scratch in NumPy** to remap pixels between corresponding faces.
- Solved a per-triangle affine transform from 3-point landmark correspondences (**3-point** → **2×3 matrix**), combined with polygon-masked, piecewise Delaunay-triangle warping to compute image morphs and mean faces.

Skills

Languages: Python, SQL

Technologies: PyTorch, MuJoCo, Gymnasium, Stable-Baselines3, NumPy, dlib, scipy, Git